

Regression Analysis (II)

Midterm Exam.

October 18, 2021

1.(80) Consider a multiple linear regression $y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i$, $i = 1, \dots, n$.
(1) (40) Construct the likelihood function for $\boldsymbol{\beta}, \sigma^2$, and the transformation parameter λ under the Box-Cox transformation model, where σ^2 is the variance of the error terms under the Box-Cox transformation model.

(2) (40) Compute the MLEs of $\boldsymbol{\beta}, \sigma^2$, and λ based on the likelihood function in (1).

2.(40) Express a linear regression model representing two equations; (i) $E(Y) = \beta_0 + \beta_1 X$, $X \leq c$ and (ii) $E(Y) = (\beta_0 - c\beta_2) + (\beta_1 + \beta_2)X$, $X > c$.

3.(80) The fitted model for the 2nd order with 2 covariates is $\hat{y} = 1 + x_1 + x_2 + x_1^2 - 2x_1x_2 + x_2^2$.
(1)(40) Find a stationary point.

(2)(40) Is the fitted value at the stationary point maximum, minimum or saddle point?

4.(40) Consider a multiple linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$. Assume that $\boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, and that the prior for $\boldsymbol{\beta}$ is $N_p(\mathbf{0}, \frac{\sigma^2}{\theta} \mathbf{I})$. Show that the Bayes estimator under the squared error loss is the ridge regression estimator.

5.(120) Consider a simple linear regression model with no intercept $Y_i = \beta X_i + \epsilon_i$, $i = 1, \dots, n$, $\epsilon_i \sim N(0, \sigma^2)$.

(1)(40) Derive a test statistic for $H_0 : \beta = 0$.

(2)(40) Find a $100 \times (1 - \alpha)\%$ C.I. for a particular response $Y(x)$ at $X = x$.

(3)(40) Consider an inverse regression problem. First, find a point estimator $\hat{x}_0$ when the response y_0 is given. Also, find a $100 \times (1 - \alpha)\%$ C.I. for $\hat{x}_0$.

6.(40) By the SVD, we have $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$, where $\mathbf{X}$ is an $n \times p$ design matrix, $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices, and $\mathbf{D}$ is a diagonal matrix with singular values as diagonal components. Construct a principal component regression model based on the 1st M principal components.